Selim Amar

Contact Information

Stanford, California, USA selama@stanford.edu

Relevant Skills

Languages:	English, French
Coding:	Programming: Python — NumPy, SciPy, scikit-learn, Matplotlib
Mathematics:	PDEs, finite element methods, probability, basic deep learning

Education

Stanford University

PhD in Mathematics, (2022-Present)
Expected date of graduation: 2028

McGill University

B.Sc. in Honours Mathematics and Physics, (2019-2022); GPA: 4.0

Academic Research Experience

2022-Present — Supervised Research on Microlocal Analysis.
PhD Advisor: V. András, Department of Mathematics, Stanford University

Developed new analytic methods for non-elliptic PDEs with hyperbolic dynamics; proved Fredholm estimates for wave-type operators in the presence of normally hyperbolic trapping. Techniques involve spectral theory, functional analysis, pseudodifferential operators, and dynamics.

2021 — Peeling theorem in higher dimensions.
Supervisor: N. Kamran, Department of Mathematics, McGill University

Derived asymptotics for waves on high-dimensional spacetimes. Techniques involved asymptotic analysis, ODE theory and differential geometry.

2020 — Discrete exterior calculus.
Supervisor: G. Tsogtgerel, Department of Mathematics McGill University.

Conducted numerical experiments on the convergence of simulations of PDEs using the discrete exterior calculus. Coding done with Python.

Publications

S. Amar, Fredholm Theory of Non-Elliptic Operators in the Presence of Normally Hyperbolic Trapping, Preprint available on arXiv at https://arxiv.org/abs/2510.16642.
S. Amar, A peeling theorem for the Weyl tensor in higher dimensions, Classical and Quantum Gravity (2022). Preprint available on arXiv at https://arxiv.org/abs/2111.08638.

Relevant Talks

Fredholm Theory of Non-Elliptic Operators with Trapping (McGill University Geometric Analysis Seminar and Berkeley Spectral Theory Seminar)
Neural networks and PDEs: from physics-informed learning to unstable solutions (Student Learning Seminar on Mathematics and Computation — Stanford University)

Selected Honors and Awards

2024-Present	FRQNT Doctoral Training Scholarship Department of Mathematics and Statistics, Stanford University
2022	Gowrisankaran Prize in Mathematics and Statistics — Department of Mathematics and Statistics, McGill University
2022	Horace Watson Prize — Department of Physics, McGill University
2021-2022	James F Mathison Scholarship — Department of Physics, McGill University
2021	NSERC Undergraduate Student Research Award — Natural Sciences and Engineering Research Council of Canada